

Machine learning systems engineer building production ML services and GPU infrastructure. I built one of the largest speech-to-text systems in the world, replatformed billions of sensor readings, trained models to automate a hedge fund, and much more.

I like building foundational technology on small teams, scaling systems and optimizing performance as the company grows.

TECHNICAL SKILLS

Machine learning: Python, PyTorch, NVIDIA Triton, TensorRT-LLM, CTranslate2, NVIDIA Riva, vLLM

Infrastructure: Kubernetes, GCP, AWS, Terraform, Helm, ArgoCD, KEDA, Temporal, GitHub Actions

Services & data: FastAPI, Pydantic, asyncio, PostgreSQL, TimescaleDB, WebSockets, gRPC

EXPERIENCE

Abridge

Remote

Senior Machine Learning Engineer

Aug 2024 - Dec 2025

Machine Learning Engineer

Apr 2023 - Jul 2024

Joined a Series B healthcare startup as employee #25 and ML engineer #2; the company grew to >400 employees and a \$5B valuation

- Created and operated the in-house speech-recognition platform on GKE, transcribing **4-8 hours of audio per second** in production.
- Migrated ASR inference from **2,000+ NVIDIA L4 GPUs** running CTranslate2 to **100-200 H100s** running TensorRT-LLM for the same production traffic load.
- Deployed **real-time ASR** on GKE and H100s using NVIDIA Parakeet and Riva, delivering live in-app transcription over WebSockets and gRPC.
- Built a unified FastAPI interface across ML services, standardizing upstream integration, observability, and operational patterns.
- Introduced Temporal for complex asynchronous workflows, extending adoption beyond ML to other engineering teams.
- Engineered a custom approach to monorepo development, supporting dozens of engineers while minimizing churn and automating semantic versioning.
- Established shared GKE deployment practices with Platform using Terraform, Helm, ArgoCD, and KEDA.
- Mentored engineers across ML and other teams through code reviews and hands-on support for service architecture and deployment patterns.
- Led ML incident response and investigations across application, Kubernetes, and networking layers.

Socure

Remote

Staff Data Engineer, Data Science Enablement & Operations

Jun 2022 - Jan 2023

Most senior engineer supporting data science teams with tooling and process

- Integrated an internal service into the primary pipeline to reduce customer onboarding time.
- Wrote Python libraries to simplify Airflow pipeline development and migrated existing pipelines to new patterns.

Betterment

Remote

Lead Data Engineer

Jun 2021 - Dec 2021

Senior engineer on a centralized data engineering team owning databases and pipelines

- Rewrote core reporting pipelines in Airflow from R to Python.
- Introduced Pydantic for AWS Lambda tasks and simplified the data engineering platform's developer experience to increase contributions from other teams.

EXPERIENCE CONTINUED

Everactive

Senior Staff Engineer, Cloud Platform

Staff Engineer, Data Science

Ann Arbor, MI

Jan 2021 - May 2021

Mar 2019 - Jan 2021

First data science hire at a Series B industrial IoT startup

- Led the migration of **billions of rows of core sensor data** from OpenTSDB to TimescaleDB/PostgreSQL, simplifying the stack and improving customer-facing responsiveness.
- Served as technical lead for a company-wide API consolidation project, standardizing access to core sensor data across teams.
- Negotiated strict JSON Schema contracts with hardware teams to define interfaces for sensor data.
- Led the shift toward event-driven platform architecture on Kafka.
- Created Python libraries and AWS ECS services for automated failure detection from sensor data.
- Wrote job descriptions and conducted interviews for subsequent data science hires.

Methods Consultants

Data Science Engineer

Ypsilanti, MI

Mar 2016 - Feb 2019

First full-time hire and lead engineer at a boutique data science consulting firm

- Developed a semi-automated ML trading platform in R and Python for a mid-size hedge fund, running on GKE and Pachyderm and supporting **over \$100k in daily trading volume**.
- Analyzed **billions of geolocation pings** with R and OpenStreetMap to support expert-witness testimony in a trucking fraud case.
- Developed and taught a **week-long R and SQL workshop** for analysts at a large federal institution.
- Introduced version control with Git and GitLab and cloud computing with AWS and GCP, establishing foundations for the firm's engineering work.
- Hired and mentored junior data scientists.
- Contributed technical writing to the company blog, including [A Tour of Timezones \(& Troubles\) in R](#).

University of Michigan School of Social Work

Data Analyst (part-time)

Ann Arbor, MI

Aug 2015 - Dec 2017

Industry consultant supporting a data-driven social work research lab

- Built predictive models to classify journal abstracts, redesigned pipelines for administrative data, and created reusable lab tooling.
- Taught source control and reproducible research to analysts and the broader School of Social Work.

Barracuda Networks

Business Intelligence Analyst

Performance Analyst/Software Engineer in Test

Ann Arbor, MI

Mar 2014 - Feb 2016

Aug 2012 - Mar 2014

QA engineer turned founding business intelligence analyst at a large software company

- Created data warehouses to provide the company's first historical view of critical production data.
- Managed server infrastructure and developed R tooling for cloud capacity planning.
- Identified performance regressions and worked with engineering to determine their causes.

Michigan Tech Research Institute

Research Intern, Signal/Sensor Processing Lab

Ann Arbor, MI

May 2012 - Aug 2012

Postgraduate summer internship at a university-affiliated R&D lab

- Wrote MATLAB image-processing filters, analyzed radar data, and extended legacy C code and Bash scripts.

EDUCATION

Michigan Technological University

Bachelor of Science in Mathematics, minor in Economics

2012